

30 CHAPTERWISE IMPORTANT QUESTIONS

Science (086) | CBSE 2026-27

Acids, Bases and Salts

Formative-only topic in CBSE 2026-27: useful for school assessment; not year-end board assessed

30 original revision questions with answers: short-answer, competency, assertion-reason and long-answer practice. Set is based on the current syllabus and official question-paper style; it is not a prediction or an official CBSE paper.

Question 1

How can litmus and universal indicator distinguish an acid from a base?

Answer:

An acid turns blue litmus red and a base turns red litmus blue. A universal indicator gives a range of colours: acidic solutions are below pH 7 and basic solutions above pH 7.

Question 2

Write the reaction of an acid with a metal carbonate and name the gas released.

Answer:

Acid + metal carbonate $\rightarrow$ salt + water + carbon dioxide. For example, $2\text{HCl} + \text{Na}_2\text{CO}_3 \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$. Carbon dioxide turns limewater milky.

Question 3

Why is baking soda used in some fire extinguishers and in baking?

Answer:

On heating or reacting with an acid, sodium hydrogen carbonate releases carbon dioxide. The gas helps smother flames; in dough it forms bubbles that make the baked product rise.

Question 4

What is the central idea or skill tested in Acids, Bases and Salts?

Answer:

The central focus is: An acid turns blue litmus red and a base turns red litmus blue. This is the starting point for a strong answer from this chapter.

Question 5

Write two relevant points for a short answer based on Acids, Bases and Salts.

Answer:

First point: An acid turns blue litmus red and a base turns red litmus blue. Second point: Acid + metal carbonate $\rightarrow$ salt + water + carbon dioxide. Link the two points clearly instead of listing them without explanation.

Question 6

How does the first key idea in Acids, Bases and Salts lead to an important result or conclusion?

Answer:

Begin with the key idea: An acid turns blue litmus red and a base turns red litmus blue. Then connect it to the result: Acid + metal carbonate $\rightarrow$ salt + water + carbon dioxide.

Acids, Bases and Salts - Continued

Question 7

What evidence, calculation, incident or concept best supports the first main idea in Acids, Bases and Salts?

Answer:

Use this supporting detail: An acid turns blue litmus red and a base turns red litmus blue. A universal indicator gives a range of colours: acidic solutions are below pH 7 and basic solutions above pH 7. The relevant detail should be stated before giving the final conclusion.

Question 8

Explain the importance of the second major idea in Acids, Bases and Salts.

Answer:

Acid + metal carbonate $\rightarrow$ salt + water + carbon dioxide. For example, $2\text{HCl} + \text{Na}_2\text{CO}_3 \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$. Carbon dioxide turns limewater milky. Its importance lies in showing how the chapter develops the topic beyond a simple fact or definition.

Question 9

What common mistake should a student avoid while answering a question from Acids, Bases and Salts?

Answer:

Do not give only a conclusion. Explain the reasoning with a correct detail, such as: On heating or reacting with an acid, sodium hydrogen carbonate releases carbon dioxide.

Question 10

Compare the first and second important ideas in Acids, Bases and Salts.

Answer:

The first idea is: An acid turns blue litmus red and a base turns red litmus blue. The second idea is: Acid + metal carbonate $\rightarrow$ salt + water + carbon dioxide. A complete comparison states both the difference and the connection between them.

Question 11

Show a cause-and-effect link from Acids, Bases and Salts.

Answer:

Cause or starting point: An acid turns blue litmus red and a base turns red litmus blue. Effect or outcome: Acid + metal carbonate $\rightarrow$ salt + water + carbon dioxide. Use connecting words such as 'therefore', 'as a result', or 'thus'.

Question 12

What is the most useful sequence for explaining the main content of Acids, Bases and Salts?

Answer:

A clear sequence is: An acid turns blue litmus red and a base turns red litmus blue. Next, Acid + metal carbonate $\rightarrow$ salt + water + carbon dioxide. Finally, On heating or reacting with an acid, sodium hydrogen carbonate releases carbon dioxide.

Acids, Bases and Salts - Continued

Question 13

Give a 40-50 word answer outline for an important question from Acids, Bases and Salts.

Answer:

Start with the direct answer: An acid turns blue litmus red and a base turns red litmus blue. Add one supporting point: Acid + metal carbonate $\rightarrow$ salt + water + carbon dioxide. End with the outcome or message: On heating or reacting with an acid, sodium hydrogen carbonate releases carbon dioxide.

Question 14

Which scientific observation, equation, diagram, or principle should be written accurately in Acids, Bases and Salts?

Answer:

Use an accurate scientific statement: An acid turns blue litmus red and a base turns red litmus blue. Then link it to the observation, process, or result asked in the question.

Question 15

How can an examiner see that you understand, rather than memorise, Acids, Bases and Salts?

Answer:

Connect the idea to its reason or result. For instance: Acid + metal carbonate $\rightarrow$ salt + water + carbon dioxide. Then explain why this point matters in the chapter's context.

Question 16

State one inference that can be drawn from the main situation or concept in Acids, Bases and Salts.

Answer:

A sound inference is that on heating or reacting with an acid, sodium hydrogen carbonate releases carbon dioxide. This inference must be supported by a relevant point from the chapter.

Question 17

Frame a short competency-based response using the first two ideas in Acids, Bases and Salts.

Answer:

Apply the concept by first stating: An acid turns blue litmus red and a base turns red litmus blue. Then justify the response with: Acid + metal carbonate $\rightarrow$ salt + water + carbon dioxide.

Question 18

What would make an answer from Acids, Bases and Salts incomplete?

Answer:

An answer is incomplete if it omits the explanation or final implication. Include both: An acid turns blue litmus red and a base turns red litmus blue. Also add: On heating or reacting with an acid, sodium hydrogen carbonate releases carbon dioxide.

Acids, Bases and Salts - Continued

Question 19

Prepare an Assertion-Reason style explanation based on Acids, Bases and Salts.

Answer:

Assertion: An acid turns blue litmus red and a base turns red litmus blue. Reason: Acid + metal carbonate $\rightarrow$ salt + water + carbon dioxide. Check whether the reason actually explains the assertion before choosing an option.

Question 20

How should you justify an answer from Acids, Bases and Salts in two clear steps?

Answer:

Step 1: state the concept or event - An acid turns blue litmus red and a base turns red litmus blue. Step 2: support it with the explanation or result - Acid + metal carbonate $\rightarrow$ salt + water + carbon dioxide.

Question 21

What answer structure is suitable for a 3-mark question from Acids, Bases and Salts?

Answer:

Use three connected points: An acid turns blue litmus red and a base turns red litmus blue. Acid + metal carbonate $\rightarrow$ salt + water + carbon dioxide. On heating or reacting with an acid, sodium hydrogen carbonate releases carbon dioxide. Keep the points specific and write them in a logical order.

Question 22

What answer structure is suitable for a 5-mark question from Acids, Bases and Salts?

Answer:

Give an introduction, develop these key points - An acid turns blue litmus red and a base turns red litmus blue. Acid + metal carbonate $\rightarrow$ salt + water + carbon dioxide. - and close with: On heating or reacting with an acid, sodium hydrogen carbonate releases carbon dioxide. Add a diagram, equation, example, or textual reference where relevant.

Question 23

How can the ideas in Acids, Bases and Salts be applied to a new situation or case study?

Answer:

Identify the relevant principle first: An acid turns blue litmus red and a base turns red litmus blue. Then use the chapter's reasoning to explain the case: Acid + metal carbonate $\rightarrow$ salt + water + carbon dioxide.

Question 24

Which detail from Acids, Bases and Salts is most likely to earn an extra mark when explained correctly?

Answer:

A useful detail is: On heating or reacting with an acid, sodium hydrogen carbonate releases carbon dioxide. Explain it in context instead of using it as an isolated statement.

Question 25

Write a brief concluding statement for an answer on Acids, Bases and Salts.

Answer:

A suitable conclusion is: On heating or reacting with an acid, sodium hydrogen carbonate releases carbon dioxide. This conclusion should follow naturally from the explanation rather than introduce a new point.

Acids, Bases and Salts - Continued

Question 26

Make a three-point revision note for Acids, Bases and Salts.

Answer:

1. An acid turns blue litmus red and a base turns red litmus blue. 2. Acid + metal carbonate $\rightarrow$ salt + water + carbon dioxide. 3. On heating or reacting with an acid, sodium hydrogen carbonate releases carbon dioxide. Revise these points with the chapter's examples, terms, and practice questions.

Question 27

How would you self-check an answer from Acids, Bases and Salts before submitting it?

Answer:

Check that you have answered the exact question, used the key idea - An acid turns blue litmus red and a base turns red litmus blue. - and supported it with a reason, example, calculation, or outcome such as: Acid + metal carbonate $\rightarrow$ salt + water + carbon dioxide.

Question 28

Why does Acids, Bases and Salts remain important for revision?

Answer:

It combines factual recall with explanation and application. The main takeaways are: An acid turns blue litmus red and a base turns red litmus blue. Acid + metal carbonate $\rightarrow$ salt + water + carbon dioxide. On heating or reacting with an acid, sodium hydrogen carbonate releases carbon dioxide.

Question 29

Create a final 60-80 word answer plan for a long question from Acids, Bases and Salts.

Answer:

Opening: An acid turns blue litmus red and a base turns red litmus blue. Development: Acid + metal carbonate $\rightarrow$ salt + water + carbon dioxide. Conclusion: On heating or reacting with an acid, sodium hydrogen carbonate releases carbon dioxide. Use complete sentences and include all required evidence, steps, or references.

Question 30

What is one last-minute revision question you should ask yourself about Acids, Bases and Salts?

Answer:

Ask: 'Can I explain why this is true or important?' A correct self-answer should include: An acid turns blue litmus red and a base turns red litmus blue. and Acid + metal carbonate $\rightarrow$ salt + water + carbon dioxide.

Sources: CBSE Science curriculum 2026-27; CBSE Class X sample papers and marking schemes 2026-27; NCERT textbooks.